

# ActInf Livestream #026.1 ~ "Bayesian Mechanics for Stationary Processes"

*Livestream | 1:53:02*

hello everyone welcome to the active inference lab today it is august 10th 2021 and we're here in live stream number 26.1 discussing bayesian mechanics for stationary processes welcome to the active inference lab if it's your first time or not we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams so people can just raise their hand visually or in the jitsi if they want to speak this short link here has our calendar of live streams for the year and we're here in number 26 where today on the 10th and then next week on the 17th we're going to be continuing discussion on the bayesian mechanics for stationary processes paper and then we're going to launch into some other papers so take a look at that link if you want to learn more today in live stream number 26.1 the goal is going to be to learn and discuss thankfully and really appreciatively with some of the authors of the paper

we're going to be discussing this  
bayesian mechanics for stationary  
processes paper  
and  
asking questions writing things down we  
want to look up over the coming week and  
just seeing  
what is uh  
fun to talk about in this paper  
so we'll start with just an introduction  
and a warm-up of the non-authors and  
then we'll go to the authors for  
conveying anything that they'd like to  
just at the outset  
so i'm daniel and i'm a postdoc in  
california i'll pass it to awful  
work in the air modern collective  
intelligence  
i founded and coordinate the kerosene  
research laboratory on building  
inactive models for navigating the  
anthropocene  
thank you let's go to stephen and then  
dean  
hello welcome my name's steven syllett  
i'm based in toronto and uh looking  
forward to hearing the new developments  
in active influence this paper covers  
great and then  
dean  
good morning i'm i'm dean i'm a  
retired guy i live in calgary and i'm  
trying to  
see what uh new insights can be provided  
by all this math because it's quite  
impressive to me  
cool so on to lance and conor whichever  
one of you would like to go first and

sort of

give us the give us the big picture and  
how we got here and how you got to this  
work before we jump into the paper  
do you want to go

uh you you should go headlines right  
um so so hi everyone and thanks for  
inviting me daniel

so my name is lance and i'm doing a phd  
in math and neuroscience in london  
with uh cal pristine and greg pavliotis  
and basically i i came to writing this  
paper and doing this kind of research  
because i'm interested in the question  
of artificial intelligence

and i think

a lot of people who are doing machine  
learning are approaching the question  
from an engineering perspective  
but i'm more interested in the  
scientific perspective the scientific  
question of what intelligence is what  
intelligence is

and what do we really need to to create  
an intelligent system

and so

um

researching around what what people were  
doing i i found about i found out about  
the free energy principle

and to me was um

the most mathematical piece of work  
that i had that i had seen in  
neuroscience and it seemed like it could  
encompass many things

and he also had applications in active  
inference

to model the brain and

ungenerate autonomous agents  
and so now in my phd my goal is to  
deeply understand the free energy  
principle  
and  
to to lay out the mathematical  
foundations because i think the free  
energy principle started out as an  
intuition  
and gradually people have  
tried to to make it more formal and  
grounded in mathematics  
um and so this is what we're trying to  
do in this paper so it's not  
this paper is not the end but i think  
it's a step forward  
so i'm really excited to  
to share it and  
ex  
like talk about it with you guys  
awesome sounds really cool about  
starting as an intuition and then  
working to ground in mathematics or or  
then would it be grounded in intuition  
and then elaborated with mathematics so  
connor  
yeah thanks i'm happy to be here um and  
good morning good afternoon to some of  
you depending on the time zone  
so yeah my name is connor heinz i'm a  
phd student at the max punk institute  
for animal behavior the department of  
collective behavior in constance germany  
and i'm co-advised by  
carl fristen at ucl and then ian cousin  
who's my advisor in constance i'm in the  
u.s at the moment uh but yeah so i  
i'm also generally interested in

bayesian mechanics i think so lance and  
i started our our phds at the same time  
and ever since i found out that lance  
was also interested in kind of the  
mathematical foundations of the free  
energy principle we've been in  
discussions about bayesian mechanics for  
a long time so i was really grateful  
that he invited me to contribute to this  
paper  
and kind of it relates to my own  
interests because i actually used to be  
more involved in neuroscience now i've  
moved more onto the idea that  
any generic complex system can be  
described as engaging in a kind of  
process of approximate bayesian  
inference and so i'm exploring that in  
my own research in the context of  
collective collective behavior so in  
particular things like collective motion  
like schooling and fish and flocking and  
birds  
so this paper i think  
it gets at kind of the mathematical  
foundations of sentience and autonomy in  
all kinds of systems not just  
nervous systems so that's why the the  
framing in terms of stochastic processes  
i think is really powerful because it  
allows us to extend the framework to all  
kinds of complex adaptive systems not  
just the ones we might associate the  
predictive coding free energy  
formulation with like brains  
thanks and a really nice point that like  
there's activity in the nervous system  
but it's not moving or the subunits are

not moving around so what are the similarities and the differences or what's the framework that's going to allow us to accommodate systems that are moving around like the schools of fish flocks of bird and then systems where the topology is either fixed or slowly changing and we have to do similar kinds of dynamical analyses so also welcome dave so stephen i see your hand raised go for it and then anyone else who wants to ask a question at this stage yeah just i'd like to comment i think that was thanks for the introductions and those insights i think um you know as there's been this progression oh we've gone from bayesian approaches and how we had predictive processing approaches and active influence approach approaches it's it's nice to also think about okay once we start to generalize it can be nice to go back um to the bayesian for idea more broadly and see okay now the active influence supports all of that you could then start to open the net so to speak so i'm interested in a lot of community psychology approaches and even interactive theater and immersive theater where you're trying to work with people's sense-making so i like that re-broadening to maybe a more heuristic mindset so i thought that was interesting

cool so anyone can raise their hand um

if at any point but

let me just maybe start with our as we

tiptoe to the edge of the paper like

what brought you

either of the authors to this framing

did you find something in the junkyard

that was looked over or was there

something that was already being worked

on and you wanted to introduce a new

thread or how did it come to be composed

and worked on in this way

um

well so that there has been um

so i think kel fristen has been working

like for the past decade

trying to put

mathematics uh on the free energy

principle and trying to explain why

um

so as i mentioned before i think the

free energy principle started out as an

intuition

and i think he has been working for the

past decade

in on a mathematical theory or a

physical theory for why the free energy

principle

should hold um

as an explanation for

what the brain is doing

and so

what we wanted to do in this paper is to

um to actually take take these

developments uh that were mostly coming

from physics

um and that weren't necessarily as

general as they could be or as

mathematically rigorous as they could be  
and actually put it on a fair  
mathematical framework  
and try to explain it clearly and  
i mean as clearly as possible  
so that was the goal here  
so what do you think that intuition was  
or maybe there's a lot of ways that i'm  
sure fristen and others have said it but  
what is the intuition that at least you  
are pursuing with this work  
right um so i think  
the the intuition  
so i think it came it came from two two  
different ways um from from two  
different ideas so so the first idea is  
if you come from machine learning  
and you look at machine learning from a  
very broad perspective you look at many  
different algorithms  
and you look at what they do you look at  
many different problems and you look at  
what they do  
and it actually turns out um and i think  
that's something that people started to  
realize in the early like 2000s  
that a lot of these  
algorithms arguably all of these  
algorithms they can be seen as  
minimizing free energy  
um  
and so we without getting  
deeply into that um i think the moral of  
the story is that  
a lot of the things that working in  
machine learning  
are special cases of minimizing free  
energy so of doing some kind of base and



inference

and so many of these problems you can think of them as like special cases of base and inference

so i think if you come from machine learning

then would be quite natural to say

or at least you could say oh

approximate inference or free energy minimization

is the most general thing you can do in machine learning

and therefore since the brain engages in some kind of cognition

it seems reasonable to assume that it may be doing some kind of basin difference

an approximate base inference so minimizing free energy is the most general way to look at it so that's one way to get to a free energy principle but maybe it's not very convincing so i think

the other

the other way you could get to a free energy principle

um which is more grounded in physics and i think that was like um the content of the original papers

is that if you look at a system that persists over time

um

so a system that does not dissipate

then you can show

that it has to be optimizing something

um

and there and then you

there has to be optimizing some what

people call model evidence  
so the this system must have like uh  
implicitly or explicitly a model of its  
environment  
and it has to be optimizing in order not  
to dissipate it has to be optimizing  
that model  
and then you can show that  
um  
actually minimizing free energy enables  
you to do that  
so i think that's the the other way um  
anyway so so what we did in this paper  
is we we took this intuition  
of what it means to persist  
so so we  
there's like actually a clear way of  
thinking about those things in uh in  
mathematics and stochastic processes  
so what we do in this paper is we look  
at systems that persist over time  
and  
we also look at systems so that's the  
first assumption and then the second  
assumption  
is that the system has to be separated  
by a boundary with its environment  
and i think i mean intuitively it makes  
sense because we we're looking at  
systems that are bounded  
uh with their environment so they  
interact  
via a markup blanket but they cannot mix  
per se with the system it's not like  
milk and coffee if you wish  
it's like a person in an environment so  
it's  
well defined in space

um so these are the two things that we assume in this paper and then we show that you can actually describe the system as minimizing free energy in some way thanks for that it's interesting about the machine learning branch towards the fep and then the physics branch because of course there's also sort of the third leg of that triangle with the physics and machine learning so we go from that just a interdisciplinary connection or an interface to add in the third which is this active inference or the free energy type framings and all of a sudden some of the ideas that were a little bit hard to to map from machine learning and physics kind of come around the horn and maybe there's a way that we can see active inference playing a role there so again anyone can just raise their hand if they have a question or write a question in the live chat here's the road map so these were the sections of the paper which introduced that boundedness notion of the markov blanket discussed basic mechanics active inference and stochastic control and then had a nice discussion and conclusion with a few supplemental sections so what went into how you laid out the arguments or what sections do you think were um

condensed or which ones would you want  
to expand on more so first just to the  
authors on that and then avo your  
question

hello lance and

[Music]

wait commer

i have a question uh before we get into  
detail i'm curious whether you were  
aware

of the kind of criticism that an  
activist launched at the french  
principle in suffocation in cognitive  
science

right could you mention what what  
specific criticism we're talking about  
okay so i'm referring to the paper  
laying down a path by di paolo  
thomson and beer

um there is a lot of criticism that is  
just

completely out

unrelated to anything that uh acting  
says but the core

thing that seems relevant to me  
is the role of creation and historicity  
and

well self-creation

in cognition intelligence and life

so yeah the paper just i'm going to put  
it in the chat is laying down a forking  
path incompatibilities between an action  
and the free energy

principle so

first to the authors for any thoughts on  
that and then see stephen with a hand  
raised

right um

so so i haven't read this particular  
paper

um

so i i can i can't give you an answer  
that's specific to this particular paper

um i would say

i would say from a general point of view

so yeah there's a lot of criticism on

the free energy principle

and i think it's natural because the

free energy principle aims to be

seems to be very general it's also an

extremely ambitious

um in in regards to to the criticism so

far

it's it's either been on the on the

mathematics

and so this is something that we've

fixed with the with the current paper so

i think a lot of the mathematical

criticism in the literature

is void

i mean is rendered void by by this paper

and there are there are other um

i would say philosophical criticisms uh

on the free energy principle so it's not

something i'm

expert about because i'm i'm a

mathematician so i wouldn't be able to

to really debate about the philosophy

here

um i i so i i'm not aware of

like uh daniel you said you put the link

to the paper in the chat oh i'll put it

in this i'll put it in the jitsi chat in

just a second okay because i i meant i

noticed you mentioned the words like

creation and historicity

um i'm not aware of the particular argument but an argument that i've heard against the free energy principle that reminds me of what you just said and this is one of the more philosophical arguments not the mathematical things that lance just mentioned is the idea that the free energy principle the inference interpretation of a system with a markup blanket at steady state it rests on the idea of a steady state so basically there's an invariant measure of the system over time and when people have tried to say relate or claim that something like societies or large-scale socio-cultural or socio-technical systems are engaging in free energy you kind of have to enforce this assumption of a stationary uh a stationary steady-state distribution on the system and so when you say things like historicity i think of like revolutions or something right so that kind of things where the the kind of intrinsic nature of the system seems to shatter or fundamentally change and that would seem to question um the or violate the steady-state assumption which is kind of a baseline thing you need in order for the free energy principle maths to apply um i can respond to that argument but i don't actually know if that's the argument laid out in this paper that's

just kind of what my brain jumped to  
when you mention things like historicity  
and  
creation and destruction  
is that is that the the core of the  
argument though or  
uh so thing like a revolution is not  
necessarily a sign of historicity it can  
be a predetermined in the um  
in the formulation of the dynamical  
system but uh  
what i'm speaking about in the medical  
terms is the presence of the of  
difficulties that will be reversible if  
for example uh we have we drive on the  
uh where am i right of the road because  
uh some guy said so like 200 years ago  
and this  
does not leave the law ever this is an  
example of uh  
walking  
or path dependency or historicity  
uh so that is the kind of argument that  
di paulo and thompson and beer are  
averaging  
but it is um formally not proportionate  
because advocacy does not entail the  
absence of historicity  
it's bernoulli systems that do not show  
historicity ergodic system only do not  
show like irreversible  
transitions which is a much quicker  
condition  
but where  
there is a convergence  
to a better argument is that  
to have a space or description of a  
dynamical system you have to assume as

you assume

wait

english is broken assume that this space  
is

fixed throughout the evolution of a  
system and if we agree with an activists  
and a system biologists like kaufman and  
montebell and

metro monteville

that

core property of the integrity system is  
to uh well make sense uh to find  
observables and to create solutions to  
problems did not exist

beforehand then the um

the stat space is not fixed it's created  
along the systems evolution this is not  
something you can account with a  
stat space description of chemical  
system

uh that has a state attractive state  
distribution sorry

yeah i i think i would agree that

as it currently stands the free energy  
principle couldn't deal the math

couldn't deal with um varying state  
spaces where literally the actual

interpretations or like the the finite  
dimensionality of the system states is  
changing over time usually you assume  
some fixed integer dimension for all the  
stochastic processing stuff although i'm  
maybe lance could comment on that i'm  
sure that could potentially be extended  
to infinite dimensional like partial  
differential equation style formulations

but uh based i if i understood you  
correctly the first part of what you



said about path dependence and irreversibility that is all totally acute uh accommodated by the maths laid out in lance's paper so there's a very you know strong part of the flow that is irreversible another way of describing that as you could say there's a strong degree of path dependence in the trajectories of the system so path dependence per se is totally accommodated and easily um you know path dependent systems can fit into the free energy framework but the state space changing itself over time that's something that i haven't really even thought about um it's an interesting argument but i i would agree that i don't think the free energy principle currently is equipped to deal with like the state space itself being altered over time that's usually assumed to be fixed but i don't know maybe lance has a different opinion on that right um yeah i completely agree with conor so in this paper we don't discuss changing state spaces and i think in terms of future work that would be really really the next step um so so i think the the current formulation uh gives a mathematical framework to think about adapting systems but it doesn't um it doesn't really talk about

like environments that structurally  
change over time for example  
thanks um steven and then we can  
continue on on this  
i think this also speaks to that the  
legs of that stool that was being talked  
about earlier if you're coming from a  
math and machine learning you've got  
um  
you know you've got more  
lower dimensional approaches but they  
show the principles and then when you  
get into the inactive approaches  
they're looking more from a biological  
perspective  
so it might also be a little bit around  
the very high complexity but the my  
understanding is that  
the  
dependency and ergodicity and also the  
ability to do approximation  
science  
uh interpretations of the machine  
learning  
does start to make  
this math  
overcome some of those limitations um  
because it it has more  
i i i that's my understanding is that  
the the reliance on ergodicity is not as  
rigid now  
so  
let's  
go to the markov blanket and then return  
to some of these important distinctions  
like ergotic systems steady state  
stationary  
so i'm gonna put the uh markov blanket

up through time figure three um figure  
one has just sort of the the timeless  
the classic blanket and then figure  
three  
is adding a subscript t  
so having that  
uh those variables describe time  
dependent functions since that's kind of  
what we're talking about with the  
dynamical systems  
so maybe  
to the authors like how do you  
summarize what this graphical figure  
means what is it being what is being  
blanketed what does it mean to blankets  
and then how does that relate to steady  
state or stationary distributions which  
are in the title of the paper  
right  
so the the mark of blanket is really the  
way the mathematical way in which you  
you talk about boundaries  
um  
and so having a markup blanket simply  
means that the internal states here  
they're separated by  
they're separated from the external  
states by the blanket states  
and then  
the fact that uh the system is at steady  
state uh means that the the blanket  
persists over time  
and so this is really the two things  
that we need when we're talking about  
um a biological system  
or or a stone  
so we need we need something internal  
to the stone or the biological system

something external and then some kind of interface

and of course when you're talking about something very simple like a stone it's not very interesting because you don't have

any exchanges between what's internal and external

via the blanket but when you're talking about biological systems like like cells or or bacillus you do have exchanges of matter of energy and information and that's something that you can accommodate with the markup blanket that's

it's almost a tension in the definition of boundedness which is you said well there is a boundary but then also it's exchanging matter energy information so what

is the boundaries meaning

uh if those are allowed to pass through it conor and then anyone else

uh i'm unmuted right no no you're good

i'm good okay yeah yeah so i would uh i would say

that's a great question because

basically the the boundary we're talking about isn't when you take a boundary for like a physical system people often

make assumptions about causality so that something on one side of the blanket can't affect something on the other side of the blanket but because of this

exchange of matter and energy across the blanket the the actual definition of the markup blanket is a statistical one it just means that this boundary

is statistically in a sense segregates the internal on the outside and and that statistical separation can actually critically differ from just direct causal interactions so the actual definition of the markov blanket in a statistical sense is that the internal states when conditioned on the blanket states are statistically independent of the external states and vice versa the external states when conditioned on blanket states are statistically independent of the internal states and just saying that definition might not make the intuition any more clear but that really is all that it is um so then when we actually look at the dynamics of particular physical or systems or stochastic processes that statement cannot entail all kinds of causal graphs and all kinds of influences between those subsets um it's really the statistical independence which defines the boundary um yeah so i think that's that's kind of the kernel of the definition and for certain kinds of statistical distributions we can actually really precisely look at the structure of things like the stationary covariance which is basically the covariance matrix of the process at steady state and you can read off what systems are outside the blanket

what what subsets of variables outside  
the blanket what subsets are inside the  
blanket which subsets the variables are  
the blanket

um so it's a really precise definition

that kind of

goes beyond whatever intuitions you  
might bake into that on your own about  
causality

awesome points so survival

sorry

does this not entail that blanket if

there is any causal interaction between  
internal and external states then

blanket states would mediate this causal  
interaction because if they did not then  
you would have information that will go  
directly from external to internal or  
vice versa so um

settlement is correct then there is some  
causal sense to blankets

is it correct or not

yeah yeah definitely and that's a really  
good point

um so really what the blanket does

is uh to to separate the inside from the  
outside statistically and so what that  
means is that

any information

about the internal states that

any transmission of information on

anything about the internal states that  
influences external states has to do  
through the blanket states

um so there is a notion of causality but  
but it's like they

as you said the internal and external  
cannot affect each other directly they

have to do it through the blanket  
okay thank you  
awesome so dean  
and then stephen  
yeah this might be a strange question  
but that's fine so the blank so the  
blankets kind of we just use that as a  
way of describing the interface my  
question is and i cuz i  
most of the math that you guys use is is  
over my head but i appreciate that  
you've been able to come up with it my  
question is  
we're thinking about this metaphorically  
does the blanket  
come with its own perforations i mean we  
talk about it acting as a mediator but  
is there a math that can kind of show  
it's it's it's managing this  
information flow back and forth but does  
it already exist with these perforations  
present or  
is the how is the math going to tell us  
how that that  
medium works or is it did you guys  
already explain that because maybe i  
just didn't appreciate it when i read  
the paper  
um  
yeah that's a really good question so we  
didn't expand on that in the paper um  
and so it's good that we talk about it  
now  
so  
the definition of markup blanket  
as it is taking the paper  
allows for boundaries that are  
completely opaque so without any kind of

transmission but also boundaries that  
would let anything through

so

so it with this formalism is quite  
general um

and so far we don't make any assumptions  
about the structure of the blanket and  
so it could be very permeable not very  
permeable

you could have all sorts of things

um and and i think that that's also

really interesting because

then i mean ultimately we want a theory  
that describes biological systems and so  
the blanket states have some degree of  
permeability in

in in biological systems that we can  
actually exploit mathematically

so we don't we don't do that here here  
the blanket is

completely arbitrary and so you could  
describe a stone but it could also  
describe

maybe a brain or a rat or something like  
this

but then in the future would be really  
interesting to actually

impose some more structure on the  
blanket to see whether we can say  
anything more interesting about the  
system that we're looking at

thanks

steven

i think this is a

an area that's probably going to be  
growing a lot is how much of that  
transfer

process of information from the infinite



almost external states  
gets onto a blanket and then gets  
transferred to the internal states  
so the question is normally everyone  
talks about changing the generative  
model  
but there is obviously a question what  
what can you change in terms of the  
blanket but i think  
i don't think that's been  
fully explored yet so but i think that's  
a big area that  
i suppose could be seen as part of the  
generative model or the regime of  
attention but  
it sounds like this the mathematical  
model here is starting to be able to  
engage that more if it was expanded  
yeah let's walk through a few  
steps of the paper because that's really  
where we're we're building up this  
intuition or moving from the intuition  
and scaffolding it with the math  
so first you phrase  
the blanket as intervening states  
between internal and external states  
and defined the state space  
that those  
different internal external and blanket  
states can exist in so totally subject  
to to servals  
comments about state space  
and  
um maybe it'd be helpful to explain this  
big pie so what is the covariance  
precision  
what are we talking about with these  
matrices

what's a system with a certain type of matrix structure that might help us get a grasp on what the covariance matrix at steady state is actually representing about systems

right so so in the in our in our scenario here the the matrix  $\pi$  actually encodes everything about um the permeability of the branch of the blanket but more generally um how do the internal states interact with blanket states and how do the external states interact with blanket state

so matrix  $\pi$  really gives you um yeah the structure of the interaction and um

and yeah and so so we just simply showed that the mark of blanket condition so this boundary actually is equivalent to some kind of structure of the matrix by but then when we try to differentiate um living systems with non-living systems and then intelligent systems from non-intelligent systems then probably we could say a lot more about this matrix by

so like if there was non-zero everywhere in  $\pi$  what would that correspond to or what would it correspond to have all zeros what does a big number or small number mean in this precision matrix because this is truly a key variable in these formalisms

right

um

so so the  
so one extreme would be the matrix by to  
be the identity matrix  
and then this would mean that  
uh each state only interacts with itself  
and so that that's like kind of an  
extreme case of a markov blanket because  
you have a boundary  
but then you don't even have any  
transmission with the boundary like you  
have just three separate states external  
blanket and internal that don't talk to  
each other  
and then the  
the other extreme would be to  
um to to have a non-zero non-zero entry  
in the in the equation that we have  
there  
and then it would mean that  
internal states and external states can  
talk to each other directly without  
interacting via the blanket scene so  
then you don't have a blanket you just  
have three states that mix with each  
other  
and so  
yeah you do you lose  
you lose kind of the the basic point  
that says oh we're looking at systems  
that have a boundary  
um and and then in terms of the the  
magnitude of the entries so a big  
magnitude  
would say um  
i think i may get get it the  
wrong way wrong way around but i think a  
big magnitude is like you have a lot of  
interaction and if you have a very low

magnitude you have very little  
interaction  
okay  
great because this  
variable is important and this  
relationship between the one over the  
precision and the variance is gonna come  
up um a lot so we go from this idea of  
using a matrix to represent  
basically the statistical connectivity  
of different nodes and then hope that  
the structure of that matrix is such  
that it falls out or designing  
systems such that the structure of the  
matrix falls out into internal external  
and mediating blanket states  
and then  
you develop this notion that the  
that there's an expected  
uh internal state and external state  
that are downstream of the blanket  
states so what is happening with these  
formalisms or how can it be that the  
blanket state tells us about what to  
expect internally and externally  
um  
so because you have a because you have a  
boundary because you have these blanket  
states  
so so the whole  
um so let me give you an example  
uh if you think about a human  
then for example part of the blanket  
states would be what happens on your  
retina so the impression from the world  
that you get from in on the retina  
and so  
this is to say that if you have some

kind of sensory impression on your  
retina then this gives you some  
information about what's going on  
outside  
and so  
um if if on your retina you see kind of  
like a a certain shape maybe you you  
maybe  
the most like the thing that's going on  
in the environment that corresponds to  
that shape is a chair  
and maybe actually that shape that you  
see on your eyes like looks like a chair  
so then you would say oh the most likely  
external state  
is a chair  
um so this is all formalizing the fact  
that the blanket states  
what what you have on on your retina  
gives you information about what's going  
on outside  
and also influences what's going on in  
your brain because  
you need to process that information  
um  
and so yeah so we have a most likely  
external state and a most likely  
internal state  
any other  
um uh thoughts on this because then  
that's where we go from  
oh  
yes so connor go for it  
yeah i'll just add that uh  
it's critical that we're talking about  
expectations here so that's why  
we're  
writing down like a normal distribution

that's centered on this expectation  
because remember these are stochastic  
processes so for any given blanket state  
it doesn't mean that the internal state  
on the other that directly is downstream  
of that blanket state will always be  
that expected one but this is just the  
mean of the distribution but the  
powerful thing about this uh the  
bayesian mechanics for these uh gaussian  
processes

is that those um expected states are  
basically just linear functions of the  
blanket states just given by these sub  
matrices of the covariance and then  
their inverses and such like that so  
it's just important to know that if  
we're to pursue this example of the  
retina it's like

for any given retinal impression it  
doesn't mean there's always going to be  
the same cortical response right in  
visual cortex or what have you but  
there's going to be some distribution of  
cortical responses and what we're  
talking about right here is the mean of  
that distribution

very

good point so steven and then anyone  
else with a raised hand and then we'll  
move forward

so

from what is you're saying there with  
the

stochastic

processes is there's there's some  
transfer

in a in a sort of a random kind of

informational sense

and obviously it's lossy as it goes through and it's the ex the expectations i suppose at some level have to have have to be able to be confirmed or disconfirmed or find some sort of um

statistical correlation

to then actually use the information that has to be i suppose at some point there's just not enough there to work anything out or the model's not good enough to find it

i'll be interested in how much

do you have a sense of how much tends to get through

or like how long it takes for

information to

be processable like if that's if there's some sort of heuristic or whether that's a little bit too unrealistic

yeah that's a really great question i

mean in the in the brain there's

evidence that you know

you can get pretty complex i'm not a

vision neuroscientist but you can get

pretty deep in the cortical hierarchy

even up to the level of a decision

within you know 150 or sometimes less milliseconds

um so that's almost kind of so this is

another big important thing to discuss

about this paper is

the information processing we're talking

about here is on the level of

expectations right so if the brain is

really

processing like an immediate retinal

impression it doesn't have time to  
average across a ton of no like tons of  
parallel noise realizations  
like

and then and then use the expected value  
to actually do its processing this so  
the the brain in any given moment is  
actually just getting stochastic  
realizations and then it's using  
information processing on those  
realizations

so i would also i don't know how lance  
feels about this but i would hesitate in  
reading in like the expected  
cortical state given a particular  
blanket state i would i would with  
restraint from reading that as the  
informational currency that the brain  
actually works with  
um

so there is information uh transfer but  
it's not like the brain always has to  
work with expectations it can work with  
um stochastic samples

at any given moment but that's also like  
a controversial thing in

in neuroscience right now it's like what  
is the actual information the brain is  
working with but the fact that the  
latency on decision making can be so  
short suggests that the brain isn't  
really averaging over a bunch of  
parallel realizations of something in  
order to do a computation

another piece is that on this slide  
we're talking about the expectation in  
terms of the the mean the average value  
of a distribution conditioned upon



blanket states we're talking about  
average value  
of internal states conditioned on  
blanket states we haven't yet introduced  
this question about internal states  
expecting or making a prediction about  
external states so it's almost like  
expectation is being used um as just the  
sort of broad statistical concept that  
you can condition expectation on some  
other variables or  
the systems that we kind of care about  
and are interested in the ones where the  
internal states are acting as if they  
are  
making expectations or predictions about  
external states which is where  
it's taken in the immediate following  
section of the paper  
that's a really good point yeah  
what is the sigma  
function so i know there's some related  
um work  
uh like with miguel that we saw a little  
bit about like  
what is the sigma what kinds of  
gaps in the formalism does the sigma  
cover  
um  
how did it come to be this way with the  
mapping function and what does it  
reflect  
right um so i would say that the sigma  
is the most  
fundamental  
ingredient to get to the free energy  
principle  
because ultimately we want to be able to

save uh what we want to be able to say  
or or say  
say that happens or or doesn't happen is  
that  
the internal states have some kind of  
representation about the external states  
uh and so the way to get there is to  
actually have a sigma function  
that  
gives that maps from the  
expected internal state or expected  
cortical response  
to  
the expected external state like what's  
going on around you  
so there there needs to be some  
uh we need we need to find uh like a  
relationship between internal states and  
external states to be able to say  
anything about inference  
and so what what we do in in this paper  
is we give  
conditions  
under which um you you have this mapping  
sigma  
and then you you're able to say oh we  
have some kind of inference going on  
and  
i think in in terms of like a historical  
perspective  
so in previous papers the the sigma was  
assumed  
um and then and then miguel wrote this  
paper saying oh the sigma  
uh you need some assumptions to for the  
sigma to work it doesn't it doesn't  
always  
uh sigma doesn't always exist

and so what we do in this paper is we we  
give conditions under which  $\sigma$  exists  
and if these conditions fail and  $\sigma$   
doesn't exist  
and crucially i think  
these conditions can be verified from  
data  
great  
points here  
um the  
synchronization map and a few of these  
other pieces relate to  
the idea of an image  
and a kernel  
so  
maybe it'd be helpful to learn a little  
bit about what what are the image and  
the kernel reflecting i found one  
representation  
about mapping the the uh two different  
spaces to each other using that sort of  
kernel and image notation but when  
you're talking about um in lemma 2.1 the  
existence of the  $\sigma$  function  
entailing  
an image function that maps the internal  
to the external states like what is an  
image  
what is the difference between just the  
variable in itself versus an image of it  
or a kernel of it  
right so what we're trying to do is to  
to map the expected internal state to  
the expected external state  
and so what we mean by image in the  
first line here  
is simply that  
yeah we're we're trying to map variables

that could be your expected cortical responses so we don't care about anything that's not your expected political response where so we care about specific type of variables

um

and then we're trying to map these to expected external states so so that's the first line

and so then here we

we just say that basically the the gist of this result is that

if

if you have something that's going on in the environment let's say there's a chair in front of you and you get some kind of expected cortical response and then let's suppose there's a sofa in front of you and you get the ex the same expected political response

then you will have a problem because it means that your organism cannot differentiate between chairs and sofas because it has the same cortical response for the for the two two different external states

and so what we're saying is that we actually don't want that to happen we want our organism to be able to differentiate between different external states

and so this is

um the the condition three and the conditions for

uh they're both i mean each of them is equivalent to one and so it puts on the

mathematical footing what it means to be  
able to differentiate between  
different states in the environment

[Music]

awesome thank you um serval and then  
we'll ask a question from the live chat  
so

for this thing to work you need organism  
to be able to discriminate between any  
two states of its environment is that  
correct

um yeah so any two states that we model  
not necessarily  
um

yes i see that mapping like four b1 and  
b2 being equal to each other  
then it implies that the that the  
external predictions are going to be the  
same

yeah so if your expected cortical  
responses are the same  
then the expected states must be the  
same

um so then again okay  
this needs to be thought about in terms  
of representations

so the the external world does not need  
to be model modeled at the molecular  
level so it's like

a model about the external world and so  
maybe

in that model there there's things

yeah so so it can be like a coarse  
grained model of the external world

okay so you use the example of a chair  
in your sofa i'm pretty sure that  
most of the listening people can do the  
difference they're pretty sure that most

cats cannot  
well i said a cat i fly it's smaller  
why would a certain organism develop  
representation of certain states under  
the cringey principle  
um so that that's a really good point  
and i think to  
to address that mathematically we would  
need to  
to actually investigate  
a state space that changes over time  
to actually to actually address the  
question of how how things learn  
representations  
so that's not that's not what  
uh we haven't done that far that far yet  
but that's definitely an important thing  
to to address in the future so what we  
the the starting point here is  
you start with a steady state you start  
with a markov blanket  
and you have uh so the representation is  
already baked in in the markov blanket  
and and so in the in this result we just  
say oh if the mark of blanket looks like  
this  
then you have a good representation  
if the markup blanket is uh sort of more  
degenerate then you have a bad  
representation  
so in the in the next slide actually um  
um  
if you can put up the next slide  
um so we show that if the if the  
condition from the previous slide is not  
met  
then you then the organism would have a  
degenerate type of representation where

for example it cannot differentiate  
between cats and between uh chairs and  
and sofas  
and and that that i suppose is still  
fine uh but it's  
um  
so so yeah it would like conflate  
complete different external states  
um so in this paper we assume that this  
wouldn't happen because then it enables  
you to  
like derive results exactly  
but you could assume that um the  
organism conflates  
different things and then you would have  
some kind of semi-degenerate type of  
representation  
and then everything still carries out  
but instead of saying oh  
my organism does exact basic inference  
and it's like super accurate and so on  
uh you can only  
um you can only say oh my organism maybe  
is doing some kind of approximate  
inference that's good in some instances  
and bad in some other instances  
so i think  
um you can definitely relax relax that  
condition and sort of assess and  
directly  
how accurate is your representation  
okay  
to reformulate any representation must  
have a precursor sensory active state in  
this framework  
okay very interesting and also we're  
still in the um inference section we  
haven't introduced

policy planning or planning as inference  
so it doesn't need to be the perfect  
state estimator in order to make  
adequate policy so  
we've talked about that in many other  
situations about action and loop and  
everything so let's  
look through the paper and then as we  
build in pieces we can see where these  
these different segments come into play  
because we haven't talked about like  
policy or conditioning inference on  
policy yet so stephen and then i'll ask  
the question from the chat  
one thing that can also help think about  
expectations and simplify or or  
get away from the pressure of  
representations is  
that  
the the blanket can be seen more like or  
the perception visual perception is more  
like an impressionist painter  
painting  
dabs  
on a painting and then stepping out and  
looking at it and we're just continually  
like trying to find the textures and the  
patterns in  
that might help confirm our expectations  
rather than it being like we see  
the shape of the object it could be that  
as long as we've seen the type of  
textures  
um that we're feeling in the space  
um and that that some of that could be  
done at the retina level there could be  
certain things that just gets processed  
so i i suppose just couching that in the



inactivist

world of you know really

when we say representation that's a bit

unloaded in that world so it could be

statistically

uh

predictable in some way

yep and we've talked about

representation and some of the different

views on representation and

axial constants uh work on the the

armstis and the representation war

so blue's question before we continue on

with exploring the formalism blue asked

uh do the maths that you have been

working with allow for altering the

structure or boundary of the blanket

itself for example if i go blind that

sensory state is no longer available

so how do we deal which is a little bit

we addressed earlier but there's like

different

senses that the models can be changing

they can be within the box

changing at steady state

or there are these other types of

changes that actually change the

structure of

the whole boundary or the internal model

itself so what kinds of changes

violate the model and which kinds of

changes are allowing us to continue

using the model

so uh

at the moment um

what the what this paper accommodates

are

sudden changes in the blanket so you

could at any point change change the blanket and then see what happens so if for example if you go blind you could change the blanket accordingly and and then everything would carry out what this doesn't account for is blankets that continuously change over time so for example if you're looking at an organism throughout its lifetime the organism would after being born it would like grow and age and so on um so we do not accommodate that but we we can accommodate sudden changes in the nine kids awesome thank you and also just anytime people need to drop off is totally cool so let's take another look at that markov blanket persisting through time and maybe even return a little bit to what kinds of systems are we talking about here there was this very interesting piece in example 3.1 about describing which types of stochastic processes at the gaussian steady state p might fit under this phrasing so several things were mentioned conor how would you summarize this list or is this an exhaustive list is this just the tip of the iceberg what kinds of attributes do these examples share that allow them to fall under this framework and then what violates it and

moves us into a different space  
yeah so this is um so basically the  
the the core set of stochastic processes  
that we we deal with in this paper are  
diffusion processes  
um so that's just a particular type of  
markov process  
which means that basically the  
current moment is totally  
uh statistically dependent on the last  
moment  
um that's not to say that there  
can't exist mathematics that uh apply  
the free energy principle to other sorts  
of stochastic processes that's probably  
something that lance is also working on  
at the moment  
but um  
if i was to give like some  
i mean  
markov prostheses are very generic in  
the sense that tons of  
biological processes i think  
and non-biological processes could be  
cast as as markov processes  
and i think there's also an important  
aspect this is kind of  
touched uh on a little bit in the next  
section with integrator chains even  
though you might say oh but like humans  
have memory there's all kinds of deep  
temporal dependencies  
you can have very high dimensional  
markov processes where with with  
different  
levels of time scales but at the lowest  
level i think it's  
a safe mathematical assumption to make

that things are markov process at the most fine grained level so i think that the these kinds of systems are like pretty unrestricted in the kinds of things they can capture everything you know from financial markets are modeled as diffusion processes the orange dynamic process which is what we use in this paper is a really popular um process used to model things like uh decision making like the drift diffusion process which is used in kind of alternative force choice decision making tasks um and then things like evolutionary dynamics like genetic frequencies drifting over time are modeled as a stochastic like linear diffusion processes um but i should also say that we use orange strength basically linear diffusions in this work and that's how we get these nice linear synchronization maps the mathematics are all linear and easy but uh this also applies to non-linear um diffusion processes so where the drift term basically isn't just a linear map like a b like a big matrix like it would be for an ou process but it could also be some non-linear very you know kind of harsh function um one some some other restrictions is that the drift function so basically the flow that determines the motion of the next set of

states given the current state that mo  
that drift function has to be smooth  
that's another condition um in these  
kinds of diffusion processes  
um yeah that is i think i pretty much  
cover it there's like some technical  
things like a zigzag process i'm  
actually not sure what that is that's  
something to ask uh lance  
great thanks  
and what you said about that um the the  
multiple levels of integration with the  
integrator chains and about how that  
addresses some of the comments about  
historiosity um we can return to that  
because there's some i think it's really  
instructive so serval with a question  
and then anyone else who wants to ask  
one  
yeah i have a question which is  
mathematical in nature for a change  
um  
i don't get what this means for random  
nuclear systems to other three states if  
it is not ergotic because if it is  
ergodic then due to well erasity  
you can find a measure that says  
something about how likely i am to find  
a system in such or such  
subspace you should not have the  
hybridic conditions and you cannot  
exchange a distribution for time average  
i don't get  
where what  
where the distribution is coming from  
exactly in the case uh where the system  
is um  
kind of deterministic

at least

yeah i can't think of an example off the moment but there are processes that have an invariant measure

where the time and the space averages are not equivalent so if you just look up

non-ergodic stationary processes i think there's some examples with

like discrete switching processes like um bernoulli variables

where the the time and the space averages will actually not be equivalent so that so i think this

is

explicitly taking into account those those kinds of systems

um where the time and the space average wouldn't be equivalent so

for instance if you think about systems with multiple basins of attraction that might like stochastically jump between those two basins of attraction

the space average there at any given moment will not necessarily be um equivalent like a bunch of parallel realizations at any given moment the space average of that might be centered around one of the modes of that process

and that will actually be different than the time average which would actually give your distribution over both modes so that's just something that that jumps to mind but there are other examples of stationary processes

where the space and time average are not equivalent which is the definition of

ergodicity

okay but uh how do you define this as  
the sorry steady state distribution then  
what does it mean and what are the  
conditions for it to be defined

so a steady state distribution

basically just means that the statistic  
the sufficient statistics that define it

uh do not vary over time

um so they're invariant they're constant

basically

um so

for for a gaussian process the mean does  
not change over time

and the covariance does not change over  
time

so that's that's an example

um yeah i mean so the the definition of  
the stationary state actually if

if you get down to the fundamentals of  
it i would more say the stationary state  
is actually just a property of the flow

so the flow that the stochastic  
differential equation itself like the

ito style differential equation that's  
the most fundamental structure of any  
system

and the stationary state actually is  
just something whose gradients drive the  
flow

so rather than um

but but that's more an interpretational  
point but in terms of the actual okay

stationary i guess so steady state

okay but you have stationary i guess i  
know what stationary is and you have as  
a state which is defined if i understand  
well but distribution p

the question is how do you define a distribution of a generical system in the general case of a stationary nickel system

so the way you define it formally is you define basically the the a partial differential equation that tells you the density of the of the system at every time point and that partial differential equation is called the fock or planck operator and then you basically set the derivative of that partial differential equation equal to zero and then you solve for the density

so that's the most like minimal bare bones definition i can give you of the of the steady state is the density which is the solution to the focker plank operator when the focker plank operates derivative with respect to time zero that's the stationary state so another way of saying that is if i start the stochastic process in a bunch of different places and run it over time all those processes will converge to the solution to the focker blank operator which is the the steady state density okay and for what kind of system do we have a stable solution

for this

rather than say uh

i don't have the words original attractor something that switches and uh

i don't know goes to uh

new places like expensive process

um yeah so the precise condition there's

basically conditions on the drift that

will tell you whether the system



explodes or whether it converges so for  
instance for an OU process  
those solutions to the Fokker-Planck  
operator  
are

defined when the eigen values of the  
drift matrix are not too big for  
instance

but what the actual conditions are for  
all sorts of stochastic processes such  
that there are unique solutions to the  
Fokker-Planck operator I am I'm not sure  
I'm not an expert on

uh on stochastic processes that's  
something that Lance might be able to  
give a more rigorous definition to but  
it really depends on the drift  $u$   
basically the  $f$  function which tells you  
whether the system has an invariant  
measure or basically a unique solution  
and sometimes there's multiple solutions  
to the Fokker-Planck operator which means  
that you know the system will end up in  
one thing or another

thanks for these

helpful explanations just one comment  
before Dean's question is that sometimes  
the steady state can be at a higher  
moment so like in the example of a car  
there's like the position and then  
there's how fast it's moving and then  
etc higher and higher derivatives and so  
the integrator chains and variational  
inference allow us to find an invariance  
at a higher level of the distribution  
that even as the actual position of the  
car is changing like that a stationary  
process doesn't mean the car is not

moving it just means there's something about the car's movement that we can continue to integrate or derive towards finding a statistical invariance specifically one that's structured kind of like this where it's almost like a diffusion process or a flow across a field so those are some of the pieces that link together and so it's not just that that the position has to be well behaved it might look very non-linear but we're using certain ways of breaking down the problem so that at each level we can frame it as this like linked sequence of tractable calculations so

connor if you have anything to add on that or dean you can question yeah i think that's a really good point like uh so basically you can correct for a moving frame of reference through differentiation and that's kind of the basis of these integrator chains is something that might be a moving frame of reference if you center it by its frame of reference then it can actually look like something like a stationary diffusion process and you get around that by encoding the system rather in terms of absolute position something like the velocity or the momentum of these higher orders of motion

so yeah i just totally agree with everything you said basically and it's

like the historical um critique or  
comment like well that race car has a  
historiosity it's like it is but its  
position at the next time point is  
determined by its position and  
momentum and etc etc etc now  
and then that is happening through time  
so there is historiosity  
we're not ignoring that but we're  
setting up the problem with this like  
vector of higher derivatives such that  
it actually  
is being only determined by the next  
moment so  
absolutely dean  
well thanks for stealing all my under  
there daniel  
i basically was going to ask so  
just to make this clear in my head the  
metaphor that i drew is you've got  
classical mechanics and quantum  
mechanics in the room and then busy  
mechanics is kind of acting like the  
marriage counselor it doesn't really  
actually determine  
what what is  
the you know  
the math can't just be precise it just  
keeps the two other parties in the room  
and has them work out what's  
what's a gist and what's a reflection or  
am i  
am i kind of way  
way off in the in the in the reads here  
yep so the connection to quantum  
mechanics which was made in this one  
section of the paper this was actually  
something that was suggested by

um

by uh greg pavliotis who is uh lance's supervisor imperial i'm actually not convinced that the connection to quantum mechanics

is as fundamental like so i i basically i don't think that uh bayesian mechanics is in between classical and

um

uh classical and quantum mechanics i see bayesian mechanics

more as

the mechanics of statistical distributions or basically the moments of statistical distributions

so in that way if i was to put it in between two branches of physics i would probably put it more

at the

three-way in between classical mechanics

statistical thermodynamics and stochastic thermodynamics because it's really uniting

the um

the kinds of coarse-grained expected variables that you deal with with something like statistical thermodynamics

with the sorts of motion that you deal with in things like stochastic thermodynamics or classical mechanics so we're talking about the motion of beliefs or about probability distributions

and it gets interesting when you have a markov blanket because then you can start imbuing the system with these like

quasi-teleological  
interpretations of one thing being a  
belief about another thing  
but the bayesian mechanics per se  
actually i think it even  
i mean it's it's it's controversial  
whether you would call it bayesian  
mechanics but there is definitely this  
kind of statistical or belief-based  
dynamics that you get even in systems  
without a mark-off blanket even a  
one-dimensional diffusion process where  
there's no boundaries or anything you  
can describe that system as optimizing  
model evidence or minimizing free energy  
it's just  
it's it's unclear how where where the  
bayesian this is because  
the bayesian interpretation really comes  
in when you're conditioning something on  
observations and then you have  
uncertainty about another variable that  
you can only perceive via the  
observations so that's why the markov  
blanket i think actually is kind of  
fundamental to the bayesian slant on the  
mechanics  
but even without it you have free energy  
minimization and you have  
the evolution of beliefs or evolution of  
belief distributions it's just they're  
not they're not as  
they don't have a nice analogy with like  
representational mechanics or the things  
we're interested in in cognitive science  
or biology  
yeah that's the part where lance was  
talking about gist

and reflection and how a gist  
morphs into a reflection and that's  
i don't even know how you would put the  
math to that but because i don't know i  
wouldn't know where to begin but that's  
that's where i thought you guys were  
headed

am i

am i misled no i think that's right  
that's kind of um

it gets that kind of i think the thing  
that daniel was saying earlier where  
there's

there's the expected internal state  
given a particular blanket state but  
then there's the idea that any  
instantaneous internal state is actually  
also encoding an expected external state  
whether or not that particular internal  
state is the most likely one given the  
blanket state so i don't know which of  
those would map to gist versus  
reflection but that's basically the idea  
is that any given realization of the  
stochastic process

at any given moment even if it's off is  
kind of always parameterizing some just  
perception about something on the other  
side of the blanket so the markov  
blanket the fact that there's a  
multivariate system with these kinds of  
conditional independencies that really  
is fundamental to the interpretation of  
saying one side is doing some kind of  
just perception of the other side

[Music]

you just don't have one side or the  
other without the ability to distinguish

them which i think is a good example about how the intuition that there are separable systems is being built upon in this nexus as you described with thermodynamics and statistics and statistical mechanics and there's other approaches to delineate subsystems like synergetics trying to define subsystems geometrically or trying to define subsystems through various ways and maybe those will become integrated as well so steven and then anyone can write a question or raise their hand

okay thanks connor for that that explanation is quite useful as well that you know when you mentioned that there can be bayesian inference say on information coming in

but the blanket you know gives

it gives a way to to hold that information i suppose but it i can see it giving a much more viable route to integrate many sources

of information which might be also sort of staggered over time as they come in and they can be integrated i mean you can integrate across blankets whereas if everything's coming in as one mush of you know all data is coming in through the same

or or even if it's not directly data but if if you've only got one

causal

pathway

then you can't necessarily

i can't see a way that you could

disambiguate

and do bayesian inference

between

what we would use blankets for or you

could even say multi-sensory integration

then that becomes a physical channel so

i suppose one question is how would you

think about

different channels of information

as opposed to

different blankets in terms of bayesian

integration

yeah i think the the actual the

multi-channel or multi-sensory

integration story is pretty nicely

accommodated by this and this kind of

brings back to an earlier question that

daniel had about what is the

interpretation of that pi matrix so when

you're thinking about the system

minimizing free energy another way of

thinking about that is it's minimizing

prediction errors

for each of the sensory modalities so if

our blanket states say we have a

three-dimensional external state a

three-dimensional internal state and a

three-dimensional blanket state because

this generalizes to infinite i mean not

infinite but very high integer dimension

our three-dimensional blanket state you

could think of as like eyes ears and

somatosensation or something

and then the the distinct elements of

that pi matrix are basically the weights

that the system that the internal states

are assigning to each of those sensory

channels so if you actually rewrite the



flow of the internal states as a function of that pi matrix

you can basically see that the internal states has differential sensitivity to each of the three

blanket states in fact it's literally integrating them and the weights it uses to do that that sum or that average are provided by the elements of the pi matrix

so i think the multi-sensory integration is basically just a higher dimensional blanket state

what it doesn't get at with what i've just said is your idea of information from different channels coming in at different time lags or different delays that is what you would have to accommodate with something like an integrator chain where

one blanket state is actually a high order derivative of some other blanket state or something like that or of an external state so you can have different blanket states that from the perspective of internal states are all coming in at once but the actual blanket states themselves are like

longer scale temporal summaries of other variables one could be the instantaneous temperature the other could be the time derivative of temperature the other one could be the average temperature over an hour or something like that

also sometimes i think about it like from a data science example like imagine if we had three variables that are getting passed to us like height weight

and favorite color  
um maybe that reflects this covariance  
matrix like height and weight up there  
on the first two  
have highest correlation with themselves  
and then they have less correlation with  
each other and then  
the off diagonal is not having any  
covariance we don't get information  
about the favorite color based upon the  
other variables say so

multi-sensory

can just be different state variables  
that are getting passed through that  
vector it doesn't have to be that all  
the blanket states are of the same type  
the blanket states are just the set of  
insulating states

and they don't all have to be  
retinal cells now if you were making  
this type of a model of the retina it  
would be that way but then if you wanted  
to make a model of a person who had  
sound input light input

etc

you could have the blanket states  
reflecting multiple different types then  
you get this challenge of  
how is the brain integrating different  
lags and different precisions etc across  
different modalities so that's not the  
end of the story but

it it um

is phrased generally enough in the state  
space representation

such that we can be discussing in the  
same matrix multiple kinds of inputs

yeah exactly

yeah actually conor if you had any thoughts on this example like how should we read the you know  $\pi$  equals what and then  $q$  equals what

right yeah so this is a point where i i kind of wanted to bring this up earlier because

i think i i don't know

it would be good i good to talk to lance about this um i think he's coming to the next one so maybe he can follow up but

basically this is actually an example of a system where

causality does not imply

does basically not exactly correlate with the markov blanket so in that  $\pi$  matrix that's not actually the covariance right that's the inverse of the covariance

so that's a really important point

the inverse of the stationary covariance is not going to have necessarily zeros in all the same places that the covariance does

so you could have two variables that actually have positive or negative correlation with each other as measured by the square root of the covariance that are actually conditionally independent

so for instance in this example um the internal which is the first row or the first column

and the external which is the third row or the third column they're conditionally independent of each other which means there's a zero in the

corresponding entry one three or three  
one

so that doesn't mean that the variables  
are uncorrelated it means that if i  
condition on blanket states the middle  
row slash middle column  
then there's no more information i can  
get

from external states about internal  
states or from internal states about  
external states it's basically saying  
that the internal states get all the  
information they possibly can from the  
blanket states and the external states  
don't add anything else that's actually  
a different statement than saying that  
there's no stationary

steady-state correlation between  
internal and external

and it goes further than that then if  
you actually looked at the drift matrix  
which is the second equation here  
you can see

from the presence of uh elements in the  
1 3 and three one elements of that  
second equation that big thing that's  
like one one point five two  
there's actually a direct influence of  
internal states on external states in  
the sense that the motion  
of external states is directly  
influenced by internal states so if you  
were to integrate that stochastic  
differential equation there would be a  
causal

if you want to call it that a causal  
connection going from internal to  
external

but nevertheless because of the way that the stationary covariance is designed when you take its inverse internal is still conditionally independent of external so this is a really interesting and kind of counter-intuitive result that is talked about a little bit in miguel's paper it's also talked about a little bit in an upcoming paper that is um in review right now uh that that basically rigorously defines the conditions under which a causal connection actually results in conditional independence between two variables so that's basically what the example here is showing is that you could have internal directly influence the motion of external or vice versa but in the stationary sense they're still conditionally independent and therefore the markov blanket is still needed to mediate information between them awesome and it was we'll look forward to these papers um but that's a really great way of laying out how we build upon this idea of just bayesian variables and the graph that describes them to dynamical systems and their covariances through time and then frame that in terms of flow which can then be decomposed into this time reversible and a time irreversible component ties us into this entire flow uh areas of

thermal flow but also information flow  
and so also just conor while we have you  
let's take this next step which is  
let's partition  
the blanket states into two kinds of  
states  
sensory on the incoming and action or  
active states on the outgoing so what is  
going on with this blanket partitioning  
and  
how does it relate to active inference  
how does active inference relate to this  
bayesian graphical approach more  
generally  
yeah so um here we're basically making  
the move of taking the blanket states  
and they could be again  
of general dimension and then  
partitioning them themselves into two  
subsets of states  
uh active states and sensory states  
so if we  
if if you basically look at the dynamics  
of the system  
and you look at the expected variables  
you can say that you know the expected  
internal state given blanket states is  
doing inference about  
external states at the same time you can  
say that the expected active state  
given sensory states is in some sense  
minimizing prediction error  
so basically minimizing the the  
deviation between the expected internal  
and the expected external state so  
that's where you get along the lines of  
uh of basically an active inference  
interpretation and i think it's

figure six if i'm not mistaken where  
that

yeah or no that's predictive processing  
figure eight is where you can basically  
show that the active the uh most likely  
active state so again this is an  
expected active state is minimizing the  
sum of precision weighted prediction  
errors so the the move here is then you  
relate minimizing when you're minimizing  
um prediction errors in the context of  
inference it's like i'm doing optimal  
bayesian inference when i'm minimizing  
precision weighted prediction errors in  
the context of control it's like i'm  
doing pid control which basically  
manuel baltierra has shown this in a  
paper from a few years ago that is the  
sort of prediction error minimization so  
now the  
elements of that pi matrix become the  
weights that you  
that you associate to each variable that  
you're trying to control in like a pid  
system so pid is this kind of  
proportional integral derivative control  
mechanism  
so we don't do this in the paper but if  
you were to associate different blanket  
states with different like derivatives  
along an integrator chain like  
the the um integral the derivative and  
then the second derivative of some  
process  
and then those were your blanket states  
then you could say that the active the  
most likely active state was basically  
trying to control the integral

the  
first order and then the derivative of a  
given system  
trying to control them using a pid  
control scheme  
and where the weights that the pid  
controller is assigned to the integral  
the first order value and the derivative  
those weights would be exactly given by  
the entries of the pi matrix that  
correspond to the blanket variables  
awesome and actually the integrator  
chain idea  
really connects some dots between this  
variational  
higher and higher derivatives of  
challenging functions variational  
inference on nonlinear functions with  
planning as inference  
so again it's like going back to that  
car with the position and the momentum  
etc  
um  
there's one way of working where you  
just want to  
fit the function of the car  
using its again position and higher  
derivatives but then what does it look  
like when you need to predict the  
position of the car so that you can act  
and intervene  
and so this just connects between this  
uh  
graphical structure where you can have  
arbitrary number of layers of nodes so  
from left to right is going through time  
but then  
vertically within a column



is in a differentiation an integration  
relationship so we connect from looking  
at a vector  
of  
position and higher derivatives  
connect that to  
the kinds of systems that  
have uh blanket states  
that's just really cool to see  
um  
yeah  
so with yeah go ahead  
yeah sorry was there someone else had a  
question or go for it connor but anyone  
can raise their hand or any questions in  
the chat is awesome  
i was just going to say  
like it's essentially that it's that um  
this so if if we were to cast the  
blanket variables as these different  
higher this vector of higher order  
derivatives of a system  
just by that system conversion to its  
stationary state it would be implicitly  
if we started far away from the steady  
state it would be implicitly essentially  
controlling each of those variables to  
some desired set point so that's often  
the terminology using control theory is  
that i'm trying to keep the velocity  
within some range keep the position  
within some range keep the acceleration  
within some range so if you interpret  
the dynamics of the most likely active  
state as doing pid control basically  
what it's doing is it's punishing  
deviations around those desired set  
points with some control cost where the

control cost exactly is given by the pid weight assigned to the acceleration velocity position which is you can read off from the inverse of the stationary covariance i.e the the pi matrix

to kind of give a fun example of that um one way of looking at this figure 10 in the helmholtz supplement is kind of like there's an ink blot and so there's some forces that have it dispersing and some that have it staying together and then when you look down from the top that's like the steady state distribution of the ink plot and so that's sort of a diffusion way to look at it but let's look at this as like a physical hill so the z-score the the height is they're not not the statistical z-scores the height is like um the concentration at steady state of ink so whatever the diffusion processes are it's densest right in the middle and then it gets less so we can think about that as being like the bottom of the bowl

and then it gets higher up so when you drop the marble in the bowl it's going to on average go to the bottom of the bowl

and then if you were to think about that marble as like a control theoretic agent with its position which is initially not at the bottom of the bowl and then all of the higher order derivatives so initially if it's stopped on the side of the bowl it's like you're holding the higher derivatives to be zero then you release it and gravity takes effect etc

it would almost look like it was like a  
plane landing on the bottom of the bull  
because it would like accelerate and  
then it would slow down its acceleration  
and the higher order derivative it would  
act  
as if it were doing pid control  
to converge all of its and then diminish  
all the derivatives so that it would end  
up  
with zero  
velocity and the differential from the  
mean of the expectation of zero so  
that's like kind of an inanimate system  
but that shows how this diffusion  
flow  
field  
and then the free energy interpretation  
of kind of like the gibbs free energy  
dropping into the bowl finding yourself  
on the bottom of a physical bowl  
is also related to control theory so  
there's like  
so many of these  
important threads to connect together  
any other  
uh  
general comments otherwise i think just  
because  
we love active inference  
what is happening in figure eight like  
what is happening on the right side and  
maybe why is it not a monotonic uh  
drop in free energy which is what we saw  
in the other figures  
yes so this is um this is basically the  
demonstration of what you just said that  
on average the system um and that's a

very key point of what of uh just to add  
that on to what you said on average the  
marble will look like it's trying to get  
to the bottom of the bowl  
and minimize those deviations from the  
different set points for position  
acceleration velocity what have you so  
this is basically showing that  
so i'll just explain the figure on the  
left that the white line is basically  
the most ac the most likely active state  
which is one of the two blanket states  
given  
various settings of the sensory state so  
the white line is saying for any given  
setting of the x-axis what is the most  
likely setting of the y-axis  
but then we also show overlaid on that  
expectation which is just a linear line  
a linear function of the sensory state  
we also show an actual realization of  
the process and that's really the the  
key thing to note here there's a  
difference between expectations  
and which are derived from the steady  
state to statistics versus realizations  
of the process which are contaminated  
with all the  
nitty-gritty messy things of the real  
world like noise and solenoidal flow and  
stuff like that so this also goes to  
answer your second question about the  
plot on the right is why is the system  
not doing monotonic decrease on  
uh free energy  
and the reason is because there's  
solenoidal flow  
so if there is no solenoidal flow and

you average this descent across trajectories an individual trajectory even without solenoidal flow might still not do monotonic free energy descent because of noise so the stochastic noise you're adding to these diffusion processes can temporarily drive it up the gradients of the free energy but if i average across like a thousand trajectories it would look like a smooth descent on free energy because all the noise realizations would be averaged out and then there would basically just be a straight line going from wherever you started this ensemble of parallel simulations to the to the bowl the minimum free energy but because of solenoidal flow which is this component of the drift that drives the system along the iso contours of the probability density which is not pointing in the direction of the free energy minimum because of that even after you average across trajectories the free energy will not monotonically go down but it'll actually follow some average trajectory that has some curvature to it um so that's an actually interesting interpretation in the active inference sense because it's so showing that on the way to the free energy minimum it's not necessarily going to take the the path laid out by the most likely active state given sensory states it might actually take a more circuitous route and it will end up on the white

line by the time it's at the minimum but  
its journey to that white line on  
average and in individual realizations  
will not actually be along the white  
line um and that's a really interesting  
there's a connection there to be drawn  
to the study of  
basically efficient um  
diffusion processes that are used in  
things like  
markov chain monte carlo so people  
basically define design diffusions in  
order to accelerate convergence to  
a stationary target distribution this is  
used a lot in like  
stochastic  
sampling based approximate bayesian  
inference and there's actually a lot of  
evidence and there's mathematical proof  
behind this too that diffusion processes  
with this kind of curvature solenoidal  
flow are actually better at reaching the  
minimum and they do so faster rather  
than one that's just doing a direct  
descent  
on the log probability  
which is equivalent to the free energy  
nice thank you one one thing i'll point  
out about this figure on the left is  
that the x-axis has twice the uh  
scale as the y-axis  
so you can also this white line is like  
the  $y = x$  line like it's right at 0  
0  
is that that's the bottom of the bowl  
and then  
it's it's  
it's just stretched out a little bit in

one way but it's kind of like that's  
just the  $y$  equals  $x$  line that would be  
if the um  
that's just the one to one and then it  
starts up here with the  $s$  of  $t$  a of  $t$   
and then it kind of finds its way to the  
bottom and that connects to figure 10  
with the helmholtz decomposition where  
just like conor was saying we can  
separate out into this solenoidal which  
is going like around the isocontour  
and then this  
um straight-line component and so the  
extreme of one case um assuming that  
such a line of sight exists is like  
you're at the bottom of the mountain and  
then you just like straight line up to  
the top  
but then the kinds of systems that we're  
looking at here  
are where we can decompose their actions  
into an as if straight line that's  
always going up the maximum possible  
elevation gain to the mountain top or  
the maximum possible descent to the  
bottom of the bowl  
but also there's this wiggle with a  
solenoidal coupling that's guaranteed to  
be orthogonal to that first component  
and that was an interesting point about  
designing the kinds of diffusion  
processes that do help us  
find  
whether you phrase it as the top or the  
bottom but find the free energy optimal  
place on a landscape that might be very  
rugged so a lot of misleading  
elevation changes

hey daniel  
doesn't that just give another feather  
in the cap for  
interactionism over instructionalism  
can you maybe explain that a little more  
and then we'll see congress  
yeah i think what conor just pointed out  
was really  
critical if you're if we're optimizing  
we have to be  
willing to set up condition where  
we're not enforcing  
a linear  
path  
that we're actually allowing for the um  
what do you want to call it the  
degree of variability and that's what  
interactionism supports so  
i just i was just listening to that  
going yeah there's another  
there's another proof now we've got the  
mathematical evidence  
cool it's like work and play a little  
bit with um  
the most direct way  
to get those instructions might be with  
a little bit of interaction along the  
way they're just they're um  
inseparable in the actual realization of  
the process  
any thoughts on that connor  
yeah i think i think that's i mean i'm  
always excited when people kind of can  
connect solenoidal flow to specific  
domains i'm not sure what instructionalism  
or interactionism are but i'm  
assuming that it's basically saying that  
just a linear classic optimization



taking the path of steepest descent is  
actually kind of maybe  
counterintuitively not always the best  
option i think that's just what  
solenoidal flow captures in any  
stochastic dynamical system is the fact  
that there's some curvature  
uh to the actual dynamics of systems  
where they're not always trying to go to  
the mode of the density but they might  
actually get where you're trying to go  
by kind of taking an alternative route  
on the way there and it's just really  
cool that mathematically you can show  
that that's actually better than taking  
the path of steepest descent  
it's like the ball can't just go down to  
the bottom in a straight line then put  
on the brakes the real path takes a  
little bit of a different route so  
oh yeah oh i want to slip one more thing  
in because i i want to give steven a  
boost before he speaks  
and that is that i've heard steven a  
number of times  
mention the word trust  
and trust takes time  
and i think again that interaction and  
that that trust piece this this math  
confirms that we don't just have to  
intuit that there's actually now some  
some support for what takes time and  
what will actually optimize  
maybe the most  
sought after outcome  
not always necessarily the best but at  
least it gives you a  
higher likelihood of achieving an

outcome

because you can you can you can now say

trust

takes time but there's there's a sort of

a almost a mathematical imperative to

give more time over to something to

arrive at the best answer

sorry stephen but i just wanted to

you know sort of boost what you've been

saying a lot and i because i i agree

with it

thanks dean

stephen

yeah it's an interesting yeah i think

the trust piece and what you could call

real life i don't know if we would talk

about real life but

a couple of questions that sort of come

from that is saying is like you know if

you're talking about

um this this solenoidal flow is is that

property or that

um does that become

a more

optimal

um

way of achieving

radiant descent when there's more noise

or the more

rugged the landscape i suppose that's my

first question and the other question is

i just wanted your thoughts you

mentioned here like active states so

this idea of like

it's almost like sensory data coming in

on active

states and we often talk about active

states and action states i

what's the states of action  
that starts the organism does but i  
suppose  
internally  
you've just got the active states which  
are kind of like almost mathematical  
codependencies so i'd be interested in  
like i say the amount of noise or  
ruggedness or approximation that has to  
happen and how much the solenoidal if it  
if it really makes a difference or  
whether the math just is the same if  
it's smooth and and and  
just your thoughts on how states are  
talked about in terms of active or  
action  
oh if that's something you think about  
yeah  
um yeah so for your first point so when  
you say the ruggedness of the landscape  
you mean like it's not just a quadratic  
single bump but there might be multiple  
bumps  
yeah i'm actually not sure i i don't i  
don't have any intuitions about why what  
solenoidal flow would do in that case i  
think like daniel was kind of saying it  
might actually be better at finding the  
global optimum  
the research i'm aware of is just saying  
that for even  
unimodal distributions like this  
sometimes the solenoid flow actually  
gets you to  
the  
unique optimum faster  
even though you would think that direct  
descent should be fastest in the

presence of noise in the dynamics so not  
a rugged landscape but just the dynamics  
are perturbed by noise the solar nodal  
flow will actually get you to the  
optimum faster but in the case of the  
actual probability distribution of the  
steady state is multimodal and rugged  
itself that's a very interesting  
question i don't know  
i just don't know the research what it  
says and i personally don't have any  
intuition about whether solano flow  
would accelerate convergence to the  
global optimum  
um or not but i i i would be surprised  
if no one's researched that because it's  
a very interesting question  
and then regarding the active states  
versus action  
yeah i mean  
i guess that i haven't thought that much  
about what like what  
semantically jumps out to me or what i  
associate when i hear active state  
versus action but i guess i think of  
fluctuations in the active states as  
actions so as the active state evolves  
that's what i would call an action or an  
or a particular realization of an active  
state as an action and then the active  
state itself we just refer to as the  
random variable that's a stochastic  
process that evolves over time and we  
refer to that as the active state  
um so we have the both the  
two-dimensional in this case  
two-dimensional blanket states one is a  
random variable that we call the sensory

state another one is a random variable  
we call an active state so that whole  
stochastic process

is the active state and maybe individual  
realizations of it would be an action

thanks connor

stephen and then anyone else and then  
also in this last few minutes if anyone  
in the live chat has a question they can  
write it so stephen

thanks that that's helpful i think like  
you mentioned that i was thinking maybe  
particularly about noise so yeah that  
makes sense if there's more noise when  
there's more

rugged landscape yeah that's an  
interesting question

that maybe someone's worked on i was  
wondering what you're basically saying  
with active states is

there are more flux there's a there's a  
greater flux a bit like sensory state  
like the information coming in as is  
more fluctuating and i.e like well you  
could say lights going at billions of  
hertz you know or whatever

and then it's the the regularities are  
slowing down as it comes in the active  
states would be a greater flux  
compared to going through the blanket  
and the transferring it

to whatever

the perturbations

that are relevant to our model are on  
the outside so that's kind of  
interesting

one comment on that is um

the actual error dynamics or just

baseline dynamics of each of these kinds of states it's going to depend on the system but when we're thinking about biological systems yes the action states can be conditioned on what the system is like if there's one degree of freedom on the elbow joint then it's a dimensionally reduced in comparison to the incoming sensory data for example so active states end up kind of saving the day in a lot of ways which is why it is such a clutch advance in the field to partition the otherwise just co-mingling blanket states into this kind of incoming and outgoing because then we can assign those incoming states and the statistically causal chains coming inbound we can associate that with inference and with everything related to predictive processing and then on the outbound from the internal states going back out to the active states we can associate that with cybernetics control theory and also planning as inference so the partitioning is is a is an immensely important maneuver steven and then anyone else yeah and also the active states become a proprioceptive form of i mean really appropriate perceptive form of sensory input in a funny sort of way like in the sense that how much they change so for instance you push your i'm kind of curious where there's this blending you know you put your hand on a on a hot

on a wall and you push you're sensing  
the wall

but you're also sensing your active  
states aren't changing as much  
that makes sense because the wall is  
stopping them

there's an interest in

but that that all could be happening at  
this kind of

fluctuating speed

however like you just mentioned

the system structure will

the fact that you've got an elbow and

the fact that it allows certain things

to happen is going to make all that come  
into

context of the system you know so that's  
quite cool

yep um sorry if there's any noise there

but yes when the um person is pushing on  
the wall they might be getting a  
different than expected or less than  
expected

amount of change in their sensory input

wow it's really loud i'm just going to

view it for a second

incredible times so

um both conor and lance have uh departed

for today and i believe that we'll be

talking with at least lance next week in  
the 26.2

so in this last 10 minutes or so

let's look back over the paper and any

of the other figures and let's think

about what we want to ask or what we'd

like to explore

in that dot too

so

who has any thoughts of this  
i got i have a thought  
i didn't raise my hand sorry um  
at some point the the 60 year old me  
would like  
some of this be to be brought down to  
six-year-old me in terms of the  
sophistication of the math and sort of  
maybe  
maybe walk through some more of those um  
the ability to be able to explain to  
somebody that there isn't that there is  
a an underpinning of math that explains  
what this medium is about and why  
things pass through it and and  
and necessarily allow these two  
differentiated states to work  
i know that i know that they have to be  
able to explain it in  
in on the level because it's it's  
complexity so they have to be able to  
use a certain sophistication of math but  
if there's some way for them  
to be able to bring that down to the my  
pedestrian level my six-year-old level  
that would be fantastic because then  
i think it would be a lot easier for  
people to climb on board with the idea  
that  
some of this some of this  
bayesian mechanics stuff  
um doesn't have to remain remain elusive  
or out there but it's actually  
something useful that we can incorporate  
into a lot of the practices that we  
would like people to adopt  
thanks  
dean agreed good thing to think about



and also dave wrote in the chat we seem to be convinced that some kinds of partitioning simplify analysis great point so um internal external blanket is not the only partitioning of how you can partition nodes in a bayesian network and sense an action isn't the only partitioning that is possible for blanket states so different ways of partitioning the problem are going to variously set it up in a way that's going to be confusing or impossible to solve or challenging to solve or potentially easy to solve and sort of the exemplar of the easy problem to solve computationally not the only example but just one example is like again the ball rolling to the bottom of a hill smooth ball perfect gravity convex optimization it's very easy if you know that the shape of the bowl is a quadratic function then all you have to do is just yep go downhill wherever you are really simple heuristic algorithm for getting to the bottom of that quadratic bowl and then we look out on the ruggedness and on the changingness of the fitness landscapes or of the free energy landscapes for real systems and the entire question is again how can we partition nodes or take the derivative of sets of nodes or of different random variables so that we

can make that rugged landscape and  
transform it or at least act on a smooth  
landscape

so that's sort of the broad theme is can  
we just

cut things up

and reframe them

so that our challenge is more like the

marble going down the bowl

rather than bushwhacking you know and

we're just totally lost and we don't

even have a sense of whether we're going

up or down and all of our motion is for

naught so stephen and then anyone else

yeah i suppose in a way our cognition is

trying to help us make things look more

and more like things are rolling down a

bowl

and it's a smoother more

clear picture of the world you know as

we get older so it'd be interesting

maybe to ask a couple more questions

about how noise um

i know

how noise i mean often they use the

infomax

or softmax the infomax function i think

in a lot of active inference modeling

you know to add

some noise

or to add some distributions so i'd be

interested in how

where does the covariance

break down where does it

start to become more coherent smoother

and just their thoughts on that seems to

be quite

that seems to be relevant

beyond  
this particular paper thanks dean  
yeah one last thing  
so  
to me  
what i see is is  
the math now  
on paper talking about possibility not  
talking about  
certainty i still think that what this  
does is it opens up  
the possibility this this map opens up  
the possibility as opposed to arriving  
at a certain outcome i think all it does  
is open up that space that's what that  
markov blanket to me essentially does is  
it keeps the door open  
i know it's described as a as a blanket  
but i honestly think it what it does is  
it it creates an openness for more math  
to be able to flow in and so  
just i know they're not philosophers but  
i'd like to hear some of their  
philosophy around  
what that what that does because i think  
if people want to say oh well your math  
doesn't answer all the all the all the  
questions  
fine but it does open up a lot of things  
that previously we couldn't even say  
we're open to your point  
so i'd like to find out what their  
philosophical  
positions are now that they've been able  
to  
sort of bring this math to a place where  
it does  
prove out

cool dean thank you and yes what you  
said there about like  
uh from certainty into uncertainty and  
of course always leaving that room for  
the interaction  
it puts everything  
on the the uncertainty  
and the precision are there two sides of  
the same coin the precision is the  
inverse covariance so if two things are  
perfectly covariate there's like a one  
or a high number in that matrix  
then your precision is high  
um uh so  
or  
just  
it's in the math but yes if they're  
variable with respect to each other then  
knowing one doesn't tell you about the  
other if they're not variable with  
respect to each other then knowing one  
gives you high precision on the other  
and it puts all kinds of state  
inferences  
on this continuum of precision so  
pretty awesome conversation really  
helpful that we were able to speak with  
the authors and yeah next week in 26.2  
we'll look forward to maybe seeing some  
more of you on the stream  
having any people submit questions if  
they want to ask something or  
just have us  
digest it or  
modify a question at all  
but as you said very  
important contributions and scales  
beyond the paper so thanks again

everybody and we'll see you next week

thanks daniel bye thanks bye